Use of Optical Motion Capture in Sports Science: A case study of Golf Swing

Siriporn Noiumkar
Department of New Media
Faculty of Informatics, Mahasarakham University
Kantarawichai District, Maha Sarakham, THAILAND
e-mail: siriporn.p@msu.ac.th

Suwich Tirakoat
Department of New Media
Faculty of Informatics, Mahasarakham University
Kantarawichai District, Maha Sarakham, THAILAND
e-mail: suwich.t@msu.ac.th

***Abstract*— The research aimed to explore and demonstrate the useful technique in creating the motion picture of golf swing movement for golf beginner utilizing the motion capture tool. The sample consists of three experts chosen by purposive sampling; while, the research tools applied in this research includes: 1) the golf swing movement and gesture for golf beginner demonstrated by the motion capture tool; and 2) the questionnaire with five-level rating-scale on the proper movement of golf swing movement which was verified by the experts for its reliability. In particular, the research outcome indicated that the creation of the motion picture of golf swing for the beginner comprises of 3 major parts as follows: 1) the motion capture tool setup, suitable and sufficient space for working, camera positioning, and the right number of reflex spots on the actor's body; 2) the golf swing movement capture divided into 3 patterns by the number of the reflex spots and each pattern was done for at least 5 times by golf professionals for the movement capture; and 3) the three 3D characters and motion picture in which the researchers created with similar ratio of real human and applied the captured movement data to construct the animation character skeleton before obtaining three sets of the golf swing animation divided by the number of reflex spots. Surprisingly, the research revealed the golf professionals' suggestion that most of the current movement of golf swing has been performed in improper manners. In this regard, this research suggested that to demonstrate the proper golf swing movement for golf beginner, the key factors was not only the number and position of reflex spots on the actor's body but also the certain regulation in utilizing the motion capture tools, and that the 3D characters, in most cases, should be created with similar ratio of body as a real human.**

Keywords- Motion Capture, MOCAP, 3D Character Animation, Golf Swing, Sports Science

I. INTRODUCTION

Motion capturing is the technique used to create the character motion or animation which style of motion can record from human or talent movement. The reason of animation's movement looks like human that is the motion capture systems (MOCAP) make movement data from tracking the sensors which are attached on the actor body, then we insert the movement data to 3D character. Normally, MOCAP is the best technology in animation and movie industry such as 3D cartoon animation, special effect in movies, 3D game, and so on.

Sport is the activity with rich movement, personal skills and take much time to training. Then, technology to training is indispensability. Golf is the popular sport and difficult to play as well because of the complexity and preciseness of the body and club movement. For this reason, the authors examined the application of the MOCAP to capturing the golf swing and check the fit between 3D character animation which made from the captured data and real player swing. The result of this study will show the performance of the optical MOCAP and Sports Science application.

II. RELATED TOPICS

A. Key Terms

1) Motion Capture

Motion capture [1] is the recording of the data or information of the human or talents movement. The data captured can be as general as the simple position in space or as complex as the face or muscle masses. At the first idea of application of MOCAP is for mapping human movement onto cartoon character for making animated film. Nowadays, it has a wide variety of applications in different fields and purposes, including animation and game development, life science, military and virtual reality/simulation. Motion capture technology can be classified into three types as mechanical, optical, and electromagnetic (magnetic) [2 - 4] as shown in Figure 1. Each type has different specific and benefits as follows;



Figure 1. MOCAP systems.

a) Mechanical MOCAP System: the actor wear an articulated device suit that consists of straight rods (like a skeleton) and potentiometers (sensors in each joint of body). This system looks like exo-skeleton. Advantages of this

978-0-7695-5133-3/13 $26.00 © 2013 IEEE
DOI 10.1109/ICICM.2013.58

Authorized licensed use limited to: Mahasarakham University provided by UniNet. Downloaded on November 02,2026 at 06:49:59 UTC from IEEE Xplore. Restrictions apply.

system are: no inference from light and magnetic or electrical, real time working, and large capture range. While, limitation of movement is the disadvantage.

b) Optical MOCAP System: It is based on high contrast video imaging of reflective markers which are attached to the body or object which motion records. Then, this system consists of 3 main equipments; as high speed cameras, network, suit and reflective (passive) markers. Advantages of this system are unlimited of subject and feel free to movement, high capturing rate, a large number of markers and subjects, and the size of capturing volume. However, there are some limits such as extensive post-processing, real-time visual feedback and light inference.

c) Magnetic MOCAP System: Sometimes is called as magnetic trackers. The actor must wear an array of magnetics suit in which 12 to 20 tracking sensors are placed. The advantages are position and orientation are available without post-processing, real-time feedback, multiple actors. But, inference of magnetic and electrical is disadvantage of this system.

2) The Optical MOCAP and Animation Works

There are 3 steps of 3D character animation making by using the optical MOCAP; such as 1) system preparation 2) motion capturing and 3) post-processing and making animation. This topic illustrates the use of MotionAnalysis™ [5] optical MOCAP system which is used in department laboratory.

a) System Preparation: The steps in the preparation of the system consists of set up all of the cameras that cover the capturing volume, connect all cameras via intranet to computer which motion capture software installed. Then, processing of system calibration started. The other side, the reflective marker set is placed on the body of the actor and the subjects. Figure 2 is shown the installation of camers (left), seed calibration (middle), and placing the markers on the parts of actor bodt (right).

b) Motion Capturing: The actor and the subject are in the volume and prompt to action as defined. At the same time, the capturing software starts to record the position of marker set which is tracked by the cameras and send the data to computer.

c) Post-processing: This step is started after step 2 is finished. The important process is cleaning the loss and noise data of movement. After that take motion data to insert into 3D character and rendering the animation.

The example of motion capture and post-processing of motion capture shown in figure 3. The actor makes the movement in the step of motion capturing (left), the MOCAP operator manipulates the movement data and uses it with 3D character (middle and right).

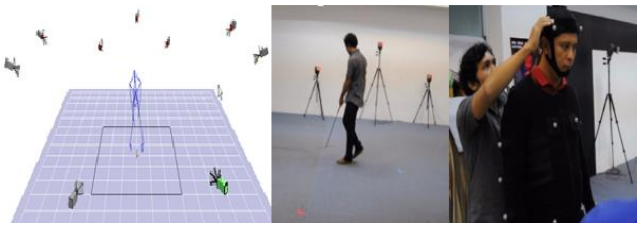


Figure 2. System preparation.

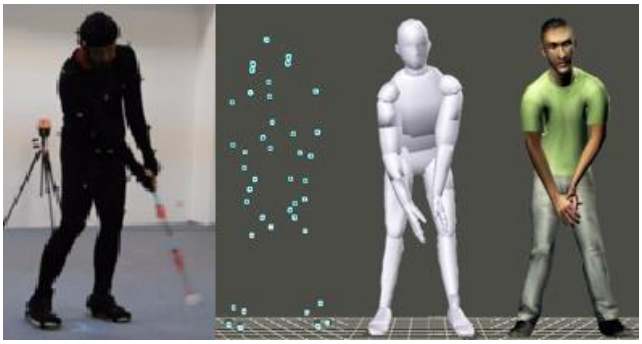


Figure 3. Capturing and making animation with MOCAP data.

B. Review of Literature

There are many researches and studies related to the application of the optical MOCAP. For example; applying motion capture technique to Thai boxing [6], this research aims to using MOCAP for record the movement of 2 Thai boxing actors and then create the animation and multimedia which is called as entertainment, the result found that the audience can gain the knowledge from the media at the good level. The application of MOCAP to sport engineering [7], this research is lab-based analysis in order to create and implement the training programs for individual improving the struggle skill by use MOCAP recording. The result found that using the proposed Judo motion capture system can significantly improve the measurement and analysis of the Judo struggle comparing to the video-based and field-based method. The instructors can also observe the positions and movements of the subject players more clearly than the other two methods. The application to medicine field [8] is studying motion capture system to human movement analysis case study of set-up defectiveness in child, the result revealed that the movement data came from MOCAP must passed some process called cleaning data to edit some loss data, and the result of comparison of motion capture data, video and mathematical analysis found that is at moderate satisfaction level.

While, the research about the optical MOCAP technique is the optimizing MOCAP [9]. This study aims to find out the small size of the optical MOCAP system as possible to work in educational institution and small and medium animation production house. The result found that 4 cameras and 24 points of marker set can create animation.

Authorized licensed use limited to: Mahasarakham University provided by UniNet. Downloaded on November 02,2026 at 06:49:59 UTC from IEEE Xplore. Restrictions apply.

III. METHODOLOGY

The optical MOCAP is the most appropriate for making the 3D character animation for entertainment industry [10-11] that is cartoon animation, special effect on the movie and games. However, this system can apply to other subjects, for example, medicine, education, science and engineering, and also sports science. This part shows the details of research methodologies which consist of objectives, scope and experimental frameworks.

A. Objective

To study and presentation of the technique in using the optical MOCAP to capturing the movement of golf swing, and to create the 3D character motion pictures that use the captured movement data.

B. Scope of Research

This study aims to find the performance of optical MOCAP with the sport, namely, golf swing was selected because of the complexity and more movement of player's body and golf club. The optical MOAP that is used to capture the movement of golf player is passive optical motion capture system, the production of MotionAnalysis (trade mark) (http://www.motionanalysis.com); 9 Eagle cameras (1.4 M and 4.0 M pixels of resolution) with capture speed at 500-2,000 frame per second, 3x5 meter of capturing area, 14 mm. of marker set size. To capture the movement of golf swing, the researchers used one actor who is professional golfer. Marker sets are attached to the main joint of actor body and club as shown in Figure 4. The final stage, there are two steps of collection of the data for finding the results of system performance, firstly, make 3D character animation with captured golf swing movement data, secondly, evaluate the correct and right of the character motion compare to the real player with the 5 rating scales questionnaire by 3 professional golfers and golf expertise. The 8 basic steps of checking full swing consists of; step 1 Set up, step 2 Back Swing-Takeaway, step 3 Back Swing-Half Swing, step 4 Top Swing, step 5 Down Swing, step 6 Impact, step 7 Follow through, and step 8 Finish.

C. The Experiment

One interested question of this test is the number and point of placing on actor body of marker set that affect to the capturing performance of the system, or not. Then, the number and placing of marker set are differentiated to be 3 types; namely normal 42 points, animation reduced 28 points and optimized 24 points [9] as shown in TABLE I and figure 4. Table I shows the number of markers attached on the various part of actor's body and the club which divided into six parts include head, shoulder and arm, body, hip, leg, and foot. Moreover, the number of marker in each section of body is significantly related to support for the movement. For example, part of the body with the most markers is shoulder and arm that shows this part is the most moving in the golf swing.

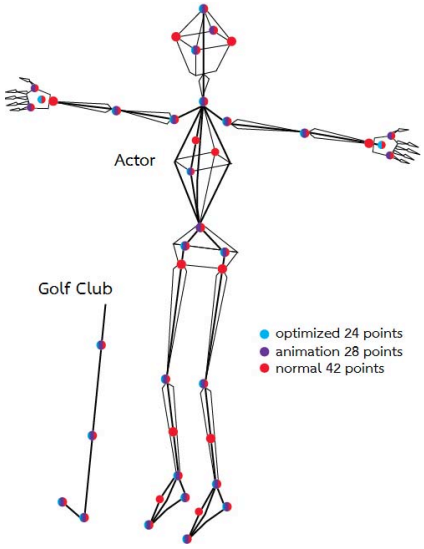


Figure 4. The 3 types of marker set.

TABLE I The number and placing point of marker set

Part of actor body where is attached marker	The number of marker set		
	Optimized 24 points	Animation 28 points	Normal 42 points
Head	2	3	5
Shoulder and Arm	6	8	12
Body	2	2	5
Hip	2	3	4
Leg	4	4	6
Foot	4	4	4
Golf club	4	4	4

IV. THE RESULTS

The researchers are made 3 animations of golf swing from variety of number of markers which shown in figure 5. Table II shows the arithmetic mean ($\bar{x}$) and standard deviation (S.D.) of scores of correct or right between 3D character animation and real player which were collected from professional and expertise in golf.

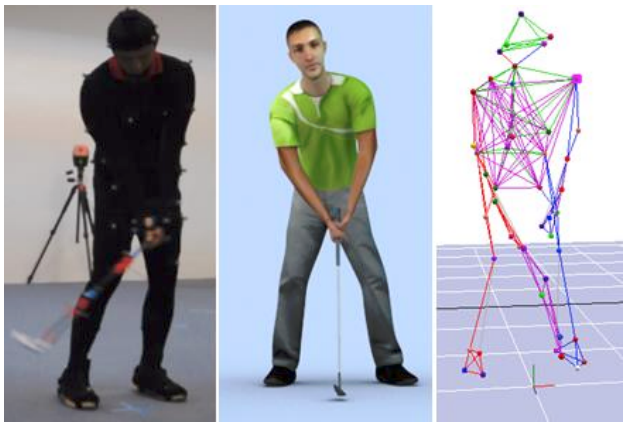


Figure 5. 3D character animation (middle) made from mocap data (right).

Authorized licensed use limited to: Mahasarakham University provided by UniNet. Downloaded on November 02,2026 at 06:49:59 UTC from IEEE Xplore. Restrictions apply.

TABLE II The level of quality of 3D charater animation

Step	Optimized 24 points		Animation 28 points		Normal 42 points	
	$\bar{x}$	S.D.	$\bar{x}$	S.D.	$\bar{x}$	S.D.
1. Set up	2.67	1.53	2.67	1.53	2.33	1.53
2. Back Swing-Takeaway	3.67	0.58	3.67	0.58	2.00	1.00
3. Back Swing-Half Swing	3.33	1.15	3.67	0.58	2.67	0.58
4. Top Swing	3.00	1.00	3.00	1.00	2.33	0.58
5. Down Swing	3.33	1.15	3.00	1.00	2.00	0.00
6. Impact	3.33	1.15	3.00	1.00	2.33	0.58
7. Follow through	2.67	1.15	3.00	1.00	2.00	0.00
8. Finish	2.67	0.58	2.67	1.53	2.00	1.00
Average	2.67	1.04	3.08	1.03	2.21	0.25

*meaning of mean score of level of correct and right
4.50 – 5.00 means the most level
3.50 – 4.49 means more level
2.50 – 3.49 means moderate level
1.50 – 2.49 means less level
1.00 – 1.49 means the least or incorrect level

From TABLE II, it is inferred that 3D character animation which is made by motion data of optimized 24 points and animation 28 points and real player are fit at the same level, however, it is at moderate level (2.50 <= <=3.49). While, 42 points of marker set made performance of 3D animation of golf swing is at less correct. Moreover, the data in TABLE III shows the result of statistical compare means score between 3 types of marker set with one-way ANOVA and post hoc test with Scheffe and LSD, it is found that all types of marker set gained mean score of correct of comparison of 3D character animation and real play are different with the statistical significance level at 0.05:

TABLE III The result of comparison of types of marker set

Score	Sum of Square	df	Mean Square	F	Sig.
Between Group	4.083	2	2.042	16.869	0.000*
Within Group	2.542	21	0.121		
Total	6.625	23			

V. CONCLUSION AND DISCUSSION

In this research, the authors need to find out the answer of the level of performance of the optical MOCAP apply to complexity and reality movement like sport. It is found that the optical motion capture can capture the movement of the sport at the moderate level when compared with mechanical and magnetic type of MOCAP. Because of the markers or sensors that sent the position of movement are fixed in the actor suit or exo-skeleton. This reason, the marker sensors cannot move with actor movement. However, the optical MOCAP is the best applying technique in the field of entertainment industry [10-11]. The result is not different from the research of application of the optical MOCAP to medicine field [8]. The supporting reasons are 1) the point of marker set place on actor body are more flexibility than other types of MOCAP, and 2) the skeletal system of 3D character may not fit to actual player or actor, it affect to the motion of 3D character animation which made from MOCAP data and character is not smooth and nature.

ACKNOWLEDGMENT

The authors are grateful to the authorities of the Faculty of Informatics, Mahasarakham University, for funding the research project in the year 2012. Thanks to the comment and suggestion made by the experts and professional golfers namely Assoc. Prof. Suracha A-mornpan, Mr. Wanchai Udomsri and golf actor, Mr. Suttipong Wiboonwiriyakul (called Pro JIP). Thanks are do to MOCAP, and animation teamwork, Mr. Apisit Sayapat, Mr. Aphichit Kongkerd and Mr. Nattakon Wisetwongsa. Thanks to the paper reviewer, Prof. Dr. Ramesh Babu Beeraka.

REFERENCES

[1] Sturman, David J, "A Brief History of Motion Capture for Computer Character Animation, " Retrieved May 10 ,2013 From http://www.siggraph.org/education/materials/HyperGraph/ animation/character_animation/motion_capture/history1.htm, 1999.

[2] M. Furniss, "Motion Capture," Retrieved May, 1 2010 From http://web.mit.edu /commforum/papers/ furniss.html, 1999.

[3] M. Kitakawa and B. Windsor, "MoCap for Artists : workflow and techniques for motion capture," Massachusetts, USA. : Elsevier, 2008.

[4] S. Dyer, J. Martin, and J. Zulauf, "Motion Capture White Paper," Retrieved May, 25 2013 From <ftp://ftp.sgi.com/sgi/A%7CW/jam/mocap/MoCapWP_v2.0.html#HDR4> December 12, 1995.

[5] Motion Analysis Corporation, "Evart 5.0 user's Manual," Santa Rosa, CA, USA. 2006.

[6] S. Phunsa, N. Potisarn, and S. Tirakoat, "Edutainment – Thai Art of Self-Defense and Boxing by Motion Capture Technique," in International Conference on Computer Modeling and Simulation, 2009. ICCMS '09, 2009, pp. 152–155.

[7] S. Thanaporn, P. Kanongchaiyos, and C. Tangmenee, "Using Motion Capture for Analysis Struggle of Judo," Journal of Engineering, Vol. 2, no. 2, pp. 49-68, 2010.

[8] S. Krudpan, K. Hensangwilai, and A. Yanrekay, "The Using of Motion Capture System to Studying in Human Movement case study of Set-up Defectiveness in Child, in the 4th Chiang Mai University Academic Conference, 2010.

[9] S. Tirakoat, "Optimized Motion Capture System for Full Body Human Motion Capturing Case Study of Educational Institution and Small Animation Production," in 2011 Workshop on Digital Media and Digital Content Management (DMDCM), 2011, pp. 117–120.

[10] C. Bregler, "Motion Capture Technology for Entertainment [In the Spotlight]," Ieee Signal Process. Mag., vol. 24, no. 6, pp. 160–158, 2007.

[11] M. Z. Patoli, M. Gkion, P. Newbury, and M. White, "Real Time Online Motion Capture for Entertainment Applications," in 2010 Third IEEE International Conference on Digital Game and Intelligent Toy Enhanced Learning (DIGITEL), 2010, pp. 139‑145.

Authorized licensed use limited to: Mahasarakham University provided by UniNet. Downloaded on November 02,2026 at 06:49:59 UTC from IEEE Xplore. Restrictions apply.